Game Theory Exam, 2020
Model Answers

1(a) : The Nash equilibrium conditions are

$$\begin{aligned}[(1-\bar{\mu}),\bar{\mu}]S^A[1-\bar{\nu},\bar{\nu}]^T &\leq [(1-\mu),\mu]S^A[1-\bar{\nu},\bar{\nu}]^T \\ [(1-\bar{\mu}),\bar{\mu}]S^A[1-\bar{\nu},\bar{\nu}]^T &\leq [(1-\bar{\mu}),\bar{\mu}]S^A[1-\nu,\nu]^T\end{aligned} \qquad [5]$$

for all $0 \leq \mu \leq 1$, $0 \leq \nu \leq 1$. These relations imply that the function

$$\nu \to ((1-\bar{\mu})b_{11} + \bar{\mu}\, b_{21}) \times (1-\nu) + ((1-\bar{\mu})b_{12} + \bar{\mu}\, b_{22}) \times \nu$$

is minimized over $\nu \in [0,1]$ at $\bar{\nu}$. Since the above function is linear and the minimum is achieved at an interior point (because the equilibrium strategy is a 'strictly' mixed), the slope is zero. It follows that

$$\bar{\mu} = \frac{b_{11} - b_{12}}{(b_{11} - b_{21} - b_{12} + b_{22})} = \beta\,. \qquad [5]$$

1(b): If Player B chooses the first (resp. second) column and player 1 chooses the mixed strategy $(1-\mu, \mu)$, then the pay-off is $a_{11}(1-\mu) + a_{12}\mu$ (resp. $a_{12}(1-\mu) + a_{22}\mu$). The safety strategy for the first player minimizes the worst case, namely

$$\max\{\phi_1(\mu), \phi_2(\mu 0\}\,.$$

where $\phi_1(\mu) := a_{11}(1-\mu) + a_{12}\mu$ and $\phi_2(\mu) := a_{12}(1-\mu) + a_{22}\mu$. [4]

The interior minimizing μ^* is located at the 'break-point', so

$$a_{11}(1-\mu^*) + a_{12}\mu^* = a_{12}(1-\mu^*) + a_{22}\mu^*.$$

Hence

$$\mu^* = \frac{a_{11} - a_{12}}{a_{11} - a_{21} - a_{12} + a_{22})}\,. \qquad [4]$$

If the game is zero sum, i.e. $B = -A$, then

$$\mu^* = \frac{-b_{11} + b_{12}}{-(b_{11} + b_{21} + b_{12} - b_{22})} = \frac{b_{11} - b_{12}}{(b_{11} - b_{21} - b_{12} + b_{22})} = \beta\,.. \qquad [2]$$

We see that $\bar{\mu} = \mu^*$, i.e. the equilibrium and safety strategies of Player A coincide.

2(a): Assume person i logs in w.p. λ, but all other people log in w.p. $\bar{\lambda}$. Consider the scenarios:

A_1: 'person i logs in and at least one person does not log in',

A_2: 'person i does not log in',

A_3: 'everyone logs in'.

$$\text{prob}\,\{A_1\} = \lambda \times (1 - \bar{\lambda}^{N-1}), \text{prob}\,\{A_2\} = (1-\lambda), \text{prob}\,\{A_3\} = \lambda \times \bar{\lambda}^{N-1}\,. \quad [3]$$

The expected pay-off is

$$1 \times \text{prob}\,(A_1) + 0 \times \text{prob}\,(A_2) - \gamma \times \text{prob}\,(A_3) = \lambda \times (1 - \bar{\lambda}^{N-1}) - \gamma \times \lambda \times \bar{\lambda}^{N-1}\,. \quad [3]$$

Since $(\bar{\lambda}, \ldots, \bar{\lambda})$ is a symmetric Nash equilibrium, the pay-off must be minimized (for the ith person) at $\lambda = \bar{\lambda}$. Since it is a strictly mixed equilibrium (so $0 < \lambda < 1$) and the pay-off is linear, it follows the slope is zero, i.e. $(1 - \bar{\lambda}^{N-1}) - \gamma\bar{\lambda}^{N-1}) = 0$. Hence

$$\bar{\lambda} = 1/(1+\gamma)^{-(N-1)}. \quad [2]$$

Prob {Everybody connects} $= (1+\gamma)^{-\frac{N}{N-1}}$. [2]

2(b): Player A strategy space: $\{A_1 := (LR), A_2 := (RL), A_3 := (LL), A_4 := (RR)\}$

(Interpretation: (LR) means 'choose L at stage 1 and R at stage 3', etc.)

Player B strategy space: $\{B_1 := (LR), B_2 := (RL), B_3 := (LL), B_4 := (RR)\}$

(Interpretation: (LR) means 'choose L if A chooses L in stage 1, and choose R if A chooses R in first stage', etc.)

From the figure, the (zero sum) pay-off matrix can be calculated as

$A \backslash B$	B_1	B_2	B_3	B_4
A_1	2	1	2	1
A_2	1	0	0	1
A_3	2	1	2	1
A_4	3	0	0	3

[7]

This has saddlepoints at (A_2, B_1) and (A_1, B_4) with decision paths: (RLL) and (RRL). [2]

(A_2, B_1) is admissible. But (A_1, B_4) is not admissible because column B_4 is weakly dominated by column (B_1). [1]

3(a): Profits of the ith company: $x_i \times (1 - \sum_j x_j) - cx_i$.

Let $(y(N), \ldots, y(N))$ be a symmetric Nash equilibrium. Then

$\psi(x_i) := x_i(1 - x_i - (N-1)y(N)) - cx_i$ is maximized at x_i.

$\nabla_{x_i}\psi(x_i)|_{x_i=y(N)} = 0$ gives $(1-c) = (N+1)y(N)$. Hence:

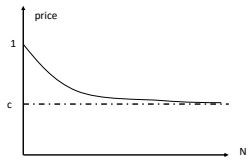
Symmetric Nash equilibrium: $y(N) = (1-c)/(N+1)$. [3]

Substituting this strategy into the profits formula gives

$$\text{Profits for company } i = \frac{1-c}{N+1}\Big(1 - \frac{N(1-c)}{N+1} - c\Big) = \Big(\frac{1-c}{N+1}\Big)^2. \quad [4]$$

At equilibrium, the price is

$$\text{Price } = (1 - Ny(N)) = 1 - \left(\frac{N}{N+1}\right)(1-c). \quad [3]$$



(Students might observe that, as the number of companies grows, the profits for each company declines to zero and the price of the product tends to the production cost).

3(b): For x_1 fixed, let $(x_2 = z(x_1), \ldots, x_N = z(x_1))$ be a symmetric Nash equilibrium for companies $2, \ldots, N$. For, $i = 2, 3, \ldots, N$,

$\phi(x_i) := x_i(1 - x_1 - x_i - (N-2)z(x_i)) - cx_i$ is maximized at $x_i = z(x_1)$.
$\nabla_{x_i}\phi(x_i)|_{x_i=z(x_1)} = 0$ gives $1 - x_1 - Nz(x_1) - c = 0$. Hence:

Symmetric equilibrium of the followers is $z(x_1) = (1 - c - x_1)/N$. [5]

The profit for company 1 (when the other companies are at equilibrium) is

$p(x_1) := x_1(1 - \frac{1}{N}x_1 - \frac{N-1}{N}(1 - c - x_1)) - cx_1$

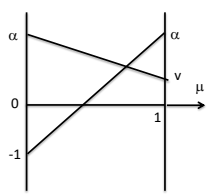
Company 1 chooses a level of production x_1^* that maximizes $p(x_1)$.

$\nabla_{x_i}p(x_1)|_{x_1=x_1^*} = 0$ gives $x_1^* = (1-c)/2$. [5]

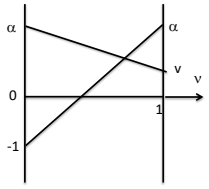
4(a): Let $((1-\bar{\mu}),\bar{\mu}),(1-\bar{\nu}),\bar{\nu}))$ be the saddlepoint in mixed strategies. By considering the 'graphical method' or otherwise, we know that $\bar{\mu}$ and $\bar{\nu}$ satisfy:

$$-(1-\bar{\mu})+\alpha\bar{\mu} = +\alpha(1-\bar{\mu})+v\bar{\mu} \text{ and } -(1-\bar{\nu})+\alpha\bar{\nu} = +\alpha(1-\bar{\nu})+v\bar{\nu}\,.$$

Solving for $\bar{\mu}$ and $\bar{\nu}$ gives $\bar{\mu}=\bar{\nu}=(1+\alpha)/(1+2\alpha-v)$ [6]



Safety Strategy for A player



Safety Strategy for B player

The pay-off (at equilibrium) $=((1-\bar{\mu}),\bar{\mu})S(1-\bar{\nu}),\bar{\nu})^T=\frac{\alpha+v}{1+2\alpha-v}$. [2]

4(b): The pay-off matrix at time $t-1$ is

$A\backslash B$	S_1	S_2
S_1	-1	$+1$
S_2t	$+1$	v_t

.

By the results of part (a) (with $\alpha=1$), the value of the game at time $t-1$ is

$$v_{t-1}=\frac{1+v_t}{3-v_t}\,.$$

Hence the value function satisfies the recursion

$$v_{t-1}=\phi(v_t),\ 0\le t\le N-1,\ \text{with } \phi(v)=\frac{1+v}{3-v}\,. \qquad [4]$$

Check the formula $v_{N-m}=\frac{m}{m+2}$ by induction:

Since $V_N=0$, the formula is correct for $m=0$. Assume it is true for general $m>0$. Then

$$V_{N-(m+1)}=\frac{1+v_{N-m}}{3-v_{N-m}}=\frac{m+1}{(m+1)+2}\,.$$

So the formula correct for $m+1$. The formula is confirmed then, by induction. [4]

(The students may alternative simply check the formula for $m=0,1,2,3$, say.)

The saddlepoint strategies for each player is (from part (a)) is

$$((1-\bar{\mu}_{t-1}),\bar{\mu}_{t-1})=((1-\bar{\nu}_{t-1}),\bar{\nu}_{t-1)}=\Big(\frac{2}{3-v_{N-m}},\frac{1}{3=v_{N-m}}\Big)=\Big(\frac{3m+3}{3m+5},\frac{2}{3m+5}\Big).$$

[4]

5(a): Bellman equation is

$$V_t(x) = \min_u \left\{ V_{t+1}\left(\frac{x+u(1+x)}{1-u(1+x)}\right) + u^2 \right\}. \qquad [4]$$

Now follow the hint that $V_t(x)$ has the structure:

$V_t(x) = p_t \left|\frac{x}{1+x}\right|^2$, $t = 0, \ldots, N$, for constants $p_0, \ldots, p_N$.

The right side of the Bellman equation is

$$\min_u \left\{ p_{t+1} \left(\frac{(x+u(1+x))/(1-u(1+x))}{1+(x+u(1+x))/(1-u(1+x))}\right)^2 + u^2 \right\}$$

$$= \min_u \left\{ p_{t+1} \frac{(x+u(1+x))^2}{(1+x)^2)} + u^2 \right\} = \min_u \left\{ p_t \left|\frac{x}{1+x}\right|^2 + 2p_{t+1}\left(\frac{x}{1+x}\right) u + (1+p_{t+1})u^2 \right\}.$$

The minimizing u is

$$u = -\frac{p_{t+1}}{1+p_{t+1}}\left(\frac{x}{1+x}\right).$$

Substituting into the right side of the Bellman equation gives:

$$p_{t+1}\left|\frac{x}{1+x}\right|^2 = \left(p_{t+1} + \left|\frac{p_{t+1}}{1+-p_{t+1}}\right|^2\right)\left|\frac{x}{1+x}\right|^2$$

We have confirmed that the proposed structure is compatible with the Bellman equation. Since the above relation is true for all x, we must have

$$p_t = \left(p_{t+1} + \left|\frac{p_{t+1}}{1-p_{t+1}}\right|^2\right), \text{ for } t = 0, \ldots, N-1, \qquad (0.1)$$

with boundary condition $p_N = 1$.

According to the principles of Dynamic Programming, the solution optimal control is given, in feedback form, by

$$u_t = -\frac{p_t}{1+p_t}\left(\frac{x_t}{1+x_t}\right) \qquad [12]$$

and the minimum cost is $p_0 \left|\frac{x_0}{1+x_0}\right|^2$. [4]